A NEW UPPER BOUND FOR THE DENSITY OF SETS IN THREE-SPACE AVOIDING UNIT DISTANCES

SERGI GALÁN

Abstract. Let $m_1(\mathbb{R}^3)$ be the supremum of the upper densities of measurable sets in $\mathbb{R}^3$ in which no two points are at distance 1. The best published bound was $m_1(\mathbb{R}^3) \le 0.1532996$, by DeCorte, Oliveira Filho and Vallentin. We prove $m_1(\mathbb{R}^3) \le 0.15158538$. We carry to three dimensions the inclusion–exclusion linear program with which Ambrus, Csiszárik, Matolcsi, Varga and Zsámboki bounded the planar density, add to it the atom δ^2 that the spectral measure of a set of density δ has at the origin, and apply it to a set of 27 points whose coordinates are rational multiples of $1/\sqrt{2}$, grown from a cuboctahedron. The proof is a rational certificate for the program, checked in exact and interval arithmetic by two programs written separately.

1. The result

A set $A \subseteq \mathbb{R}^3$ *avoids the distance* 1 if $|x - y| \neq 1$ for all $x, y \in A$. Its upper density is $\overline{\delta}(A) = \limsup_{R\to\infty} |A \cap [-R, R]^3|/(2R)^3$, and

$$m_1(\mathbb{R}^3) = \sup\{\overline{\delta}(A) : A \subseteq \mathbb{R}^3 \text{ measurable, avoiding the distance } 1\}.$$

Since a set avoiding 1 contains at most one vertex of each unit regular tetrahedron, $m_1(\mathbb{R}^3) \le 1/4$. The known upper bounds are 3/16 (Székely and Wormald [7]), 0.165609 (Oliveira Filho and Vallentin [6], by a linear program on the autocorrelation of A with extra inequalities from regular simplices) and 0.1532996 (DeCorte, Oliveira Filho and Vallentin [3, Table 2], by adding constraints of the Boolean quadric polytope on finite point sets, found by separation)[1]. Bachoc, Passuello and Thiery [2] improved the bounds in dimensions 4 to 24 but not in dimension 3. In the plane, Ambrus, Csiszárik, Matolcsi, Varga and Zsámboki [1] proved $m_1(\mathbb{R}^2) \le 0.2470$ with a linear program built from the patterns in which a random congruent copy of a finite set X meets the set A; they treated only the plane.

Theorem 1. *Every measurable set in* $\mathbb{R}^3$ *avoiding the distance* 1 *has upper density at most* 0.15158538. *That is,* $m_1(\mathbb{R}^3) \le 0.15158538$.

The proof (Section 2) is the program of [1] in $\mathbb{R}^3$, where the Bessel function J_0 becomes $\Omega(x) = \sin x/x$, together with the atom $\kappa(\{0\}) = \delta^2$ of the spectral measure, used as in [5, Prop. 3.5]; the configuration has 27 points (Section 3).

Corollary 2. *The measurable fractional chromatic number of* $\mathbb{R}^3$ *is at least* $1/0.15158538 > 6.5969$.

Proof. If $\mathbb{R}^3$ is covered by measurable sets $A_1, \dots, A_k$ avoiding 1 with weights $w_i \ge 0$, $\sum_{x\in A_i} w_i \ge 1$ at every point, averaging over a large cube gives $1 \le \sum_i w_i \overline{\delta}(A_i) \le m_1(\mathbb{R}^3) \sum_i w_i$. □

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These results were obtained with the assistance of an AI system (Claude, by Anthropic), which found the configuration, wrote the programs, ran the checks and helped to draft this note; a second program and a referee report were produced by separate AI agents. The method is that of Ambrus, Csiszárik, Matolcsi, Varga and Zsámboki [1], who used it in the plane. No mathematician has checked the note yet. The author is not a professional mathematician and submits it for independent verification.

[1] [3, Table 2] gives the previous value as 0.1645090; the tables of [6] (in both the arXiv and the journal versions) give 0.165609.

The bound $\chi_m(\mathbb{R}^3) \geq 7$ on the measurable chromatic number [6] is not changed: that would need $m_1(\mathbb{R}^3) < 1/7$. Unlike the plane [4], we know no transfer theorem that would turn the corollary into a bound for all colourings.

2. The inclusion–exclusion bound in $\mathbb{R}^3$

Let $X = \{x_1, \dots, x_n\} \subseteq \mathbb{R}^3$, $r_{ij} = |x_i - x_j|$, let E_1 be the set of pairs $\{i, j\}$ with $r_{ij} = 1$, and let $\mathcal{S}$ be the family of subsets of $\{1, \dots, n\}$ containing no pair of E_1 (including $\emptyset$). Let $\mathcal{C}$ be a list of pairs (I, J) of sequences of distinct indices of the same length with $|x_{I_a} - x_{I_b}| = |x_{J_a} - x_{J_b}|$ for all a, b. Let $\Omega(x) = \sin x/x$, $\Omega(0) = 1$.

Proposition 3. *Suppose that real numbers $\psi_0, \psi_T, m, \lambda_i$ $(1 \leq i \leq n)$, u_{ij} $(\{i, j\} \notin E_1)$ and w_c $(c \in \mathcal{C})$ satisfy*

$$\Phi(t) = \psi_0 \Omega(t) - \sum_i \lambda_i - \sum_{i<j} u_{ij} \Omega(r_{ij} t) \;\geq\; m \qquad (t > 0),$$

$$Q(S) = \psi_T + \sum_{i \in S} \lambda_i + \sum_{i<j,\ i,j \in S} u_{ij} + \sum_{(I,J) \in \mathcal{C}} w_{(I,J)} \big([I \subseteq S] - [J \subseteq S]\big) \;\geq\; 0 \qquad (S \in \mathcal{S}),$$

and $c := \Phi(0) - m > 0$, where the sums of u_{ij} run over the pairs $i < j$ with $\{i, j\} \notin E_1$. Then $m_1(\mathbb{R}^3)$ is at most the positive root of $cx^2 + mx - \psi_T$.

Proof (after [1]*).* It suffices to bound the density δ of a measurable set A avoiding 1 that is periodic for a lattice Λ [6, §2]: copies of $A \cap [-T + 1, T - 1]^3$ placed on $2T\mathbb{Z}^3$ avoid 1 and have density arbitrarily close to $\overline{\delta}(A)$ along a sequence $T \to \infty$. Let c_ξ $(\xi \in \Lambda^*)$ be the Fourier coefficients of 1_A on $\mathbb{R}^3/\Lambda$, and κ the image of the measure $\sum_\xi |c_\xi|^2 \delta_\xi$ under $\xi \mapsto 2\pi|\xi|$: a measure on $[0, \infty)$ of total mass δ (Parseval) with $\kappa(\{0\}) = |c_0|^2 = \delta^2$. The average over the orthogonal group $O(3)$ of the autocorrelation $\varphi(v) = \delta(A \cap (A - v))$ is $f(r) = \int \Omega(rt)\, d\kappa(t)$, because the average of $e^{i\langle \xi, gv \rangle}$ over $g \in O(3)$ is $\Omega(|\xi||v|)$; and $f(1) = 0$ since A avoids 1. Choose y uniformly in $\mathbb{R}^3/\Lambda$ and $g \in O(3)$ according to Haar measure, and let $S = \{i : y + gx_i \in A\}$. (The probabilities below are integrals over g of densities of atoms of the sets $A - gx_i$, which are continuous in g because translation is continuous in $L^1(\mathbb{R}^3/\Lambda)$.) Then $S \in \mathcal{S}$, $\Pr(i \in S) = \delta$, $\Pr(i, j \in S) = f(r_{ij})$, and $\Pr(I \subseteq S) = \Pr(J \subseteq S)$ for $(I, J) \in \mathcal{C}$: a distance-preserving bijection between finite subsets of $\mathbb{R}^3$ extends to an isometry $h = h_0 + b$ with $h_0 \in O(3)$, and $(y, g) \mapsto (y + gb, gh_0)$ preserves the distribution of (y, g). Reflections are allowed because g ranges over $O(3)$; 442 of the 1 078 relations of our certificate relate sets that are not coplanar by an isometry that reverses orientation. Hence $0 \leq \mathbb{E}\, Q(S) = \psi_T + \delta \sum_i \lambda_i + \sum_{i<j} u_{ij} f(r_{ij})$, and adding $\psi_0 f(1) = 0$,

$$\int \Phi \, d\kappa = \psi_0 f(1) - \delta \sum_i \lambda_i - \sum_{i<j} u_{ij} f(r_{ij}) \leq \psi_T.$$

Since $\Phi \geq m$ on $(0, \infty)$ and $\kappa(\{0\}) = \delta^2$, the left side is at least $m(\delta - \delta^2) + \Phi(0)\delta^2 = m\delta + c\delta^2$. So $c\delta^2 + m\delta - \psi_T \leq 0$; as $c > 0$ and $\psi_T = Q(\emptyset) \geq 0$, δ is at most the positive root. □

With $m = 1$, $c = 0$ this is the program of [1], with Ω in place of J_0. The atom at 0 is what lets m be negative: for X a single point, $\psi_0 = 1$ and $m = \min \Omega = -0.21723\dots$ give $\delta \leq -m/(1-m) = 0.17850\dots$, the bound of [6, (8)] for $n = 3$. The program behind Proposition 3 is linear once the condition $\kappa(\{0\}) \geq \delta^2$ is replaced by its tangents $\kappa(\{0\}) \geq 2a\delta - a^2$ (we use $a = 0.1, 0.1025, \dots, 0.1975$).

3. The configuration

All points of X_{27} have the form $q/\sqrt{2}$ with $q \in \mathbb{Q}^3$, so all squared distances are rational and we compute exactly. The first 13 points are the centre and the vertices of a cuboctahedron, that is, the points of the face-centred cubic lattice $\{q \in \mathbb{Z}^3 : q_1 + q_2 + q_3 \text{ even}\}/\sqrt{2}$ at distance at most 1 from the origin. Each further point is at distance 1 from at least three points already present (Table 1). The unit-distance graph of X_{27} has 82 edges and 44 221 independent sets.

point	coordinates	at distance 1 from
x_{14}	$(0,\ 0,\ -2)/\sqrt{2}$	$2, 5, 8, 11$
x_{15}	$(-\frac{1}{3},\ -\frac{1}{3},\ -\frac{2}{3})/\sqrt{2}$	$8, 11, 14$
x_{16}	$(0,\ -\frac{1}{5},\ -\frac{3}{5})/\sqrt{2}$	$1, 10, 14$
x_{17}	$(\frac{1}{11},\ -\frac{4}{11},\ -\frac{7}{11})/\sqrt{2}$	$1, 8, 14$
x_{18}	$(\frac{1}{5},\ 0,\ -\frac{3}{5})/\sqrt{2}$	$10, 13, 14$
x_{19}	$(0,\ \frac{1}{5},\ -\frac{3}{5})/\sqrt{2}$	$4, 13, 14$
x_{20}	$(-\frac{8}{11},\ -\frac{8}{11},\ \frac{2}{11})/\sqrt{2}$	$2, 5, 18, 19$
x_{21}	$(\frac{11}{35},\ \frac{8}{35},\ \frac{1}{7})/\sqrt{2}$	$8, 10, 20$
x_{22}	$(\frac{80}{249},\ \frac{55}{249},\ \frac{55}{249})/\sqrt{2}$	$10, 11, 20$
x_{23}	$(\frac{60}{241},\ \frac{35}{241},\ -\frac{85}{241})/\sqrt{2}$	$2, 10, 20$
x_{24}	$(\frac{8}{11},\ -\frac{8}{11},\ \frac{2}{11})/\sqrt{2}$	$5, 11, 15, 19$
x_{25}	$(\frac{7}{33},\ \frac{7}{33},\ -\frac{10}{33})/\sqrt{2}$	$2, 5, 20$
x_{26}	$(-\frac{55}{249},\ \frac{80}{249},\ \frac{55}{249})/\sqrt{2}$	$8, 13, 24$
x_{27}	$(0,\ \frac{25}{73},\ \frac{55}{73})/\sqrt{2}$	$4, 13, 20, 24$

TABLE 1. The points $x_{14}, \dots, x_{27}$ of X_{27}, in the order in which they were added, and their unit neighbours among the earlier points. Points $x_1, \dots, x_{13}$ are $q/\sqrt{2}$ for the 13 vectors $q \in \mathbb{Z}^3$ with $q_1 + q_2 + q_3$ even and $|q|^2 \le 2$, in lexicographic order ($x_7 = 0$).

4. The certificate and its verification

The congruence relations are generated as in the plane: the independent sets with at least three points fall into congruence classes (equal multisets of squared distances, then a distance-preserving bijection found by backtracking, all exact), and each class $S_1, \dots, S_k$ gives the $k-1$ relations $\Pr(S_1 \subseteq S) = \Pr(S_j \subseteq S)$.

The program is first solved in floating point, with κ supported on the grid $t \in \{0, 0.02, \dots, 60\} \cup \{60, 60.1, \dots, 100\} \cup \{100.25, 100.5, \dots, 600\}$ and the tangent cuts above; its relation multipliers, normalized to $c = 1$, are rounded to multiples of 2^{-40} and frozen. For frozen w, a small program in $\psi_0, \psi_T, m, \lambda, u$ minimizes $\psi_T - am$ (the tangent of the bound at $a = 0.1522$) subject to $Q \ge 0$ on every independent set, $\Phi(0) - m = 1$, $-\sum_i \lambda_i - m \ge 0.002$ and $\Phi(t) - m \ge 2 \cdot 10^{-5}$ on a grid. The values are rounded to multiples of 2^{-40}; m is set 10^{-5} below the minimum of Φ on a fine grid, and ψ_T is raised until $Q \ge 0$ holds exactly. The certificate is the file `m1r3_x27.json`.

Two programs, written separately, check it and prove Theorem 1:

- `check_m1r3.py` computes the unit-distance graph exactly, checks indices, that no u_{ij} sits on a unit pair and that every relation is a congruence, checks $Q \ge 0$ on all independent sets with integers over a common denominator (cross-checking their number by $I(G) = I(G - v) + I(G - N[v])$) and $c > 0$ exactly, and proves $\Phi \ge m$ on $(0, T]$ by bisection with second-order Taylor enclosures of Ω in Arb ball arithmetic and beyond T by $|\Omega(x)| \le 1/x$.
- `check_m1r3_indep.py` enumerates and counts the independent sets in its own way (three counts for $n \le 24$), evaluates Q exactly, and proves $\Phi \ge m$ with a bound t^2 near 0, third-order Taylor expansions with an explicit bound for Φ''' in between, and $|\Omega(x)| \le 1/x$ in the tail.

The facts about Ω used are: $\Omega(x) = \int_0^1 \cos(xs)\, ds$, so $|\Omega^{(k)}| \le 1/(k+1)$; $\Omega \ge -0.2173$, proved by `check_m1r3.py` on $[0, 5]$ by interval arithmetic and beyond by $|\Omega(x)| \le 1/x$. Both programs accept the certificate (in seconds) and reject corrupted copies (a lowered λ_i or ψ_T, a raised m or u_{ij}, a broken congruence, a u_{ij} on a unit pair, a moved or repeated point, $c \le 0$). A referee report, written by a separate AI agent, checked Proposition 3 against the sources and found no error; it asked for the continuity of the averaged atom densities in g and for the attribution of the previous bounds (both now in the text) and audited `check_m1r3.py`. Its own program, run on the certificate for X_{27}, recounts the 44 221 independent sets by brute force over all 2^{27} subsets, finds $\min Q > 0$ in exact arithmetic, confirms every relation (and that 442 of them reverse orientation), and finds $\Phi - m \ge 9.9 \cdot 10^{-6}$ on a dense grid (a floating-point check only).

5. How the points were found

Starting from the cuboctahedron with its centre (program value 0.178467, the single-point bound), we added one point at a time. The candidates are the points at distance 1 from three points already present (from the 20th point on, only those with coordinates in $\mathbb{Q}^3/\sqrt{2}$; the points chosen before were in it anyway); we evaluated the program for the 40 (from the 25th point on, 48) candidates of highest unit degree, ties broken by distance from the origin, and kept the best (Table 2). The gains shrink as the set grows (about $3 \cdot 10^{-4}$ per point at 24 points, $1 \cdot 10^{-4}$ for the 27th) while the number of independent sets grows by a factor of about 1.6 per point; we stopped at 27 points. The bound 1/7 that would give $\chi_m(\mathbb{R}^3) \geq 8$ is far out of reach of this search.

points	independent sets	value (floating point)
13	109	0.178467
14	144	0.177956
15	209	0.173205
16	322	0.160467
17	544	0.157160
18	882	0.156954
19	1459	0.156470
20	1741	0.155119
21	2968	0.154027
22	5018	0.152828
23	8430	0.152137
24	10175	0.151987
25	15671	0.151720
26	26117	0.151549
27	44221	0.151456

Table 2. The value of the program for the first k points of X_{27}, discretized with step 0.1 up to $t = 100$ and 0.25 up to 400 (a little below the value of the finer program used for the certificate).

References

[1] G. Ambrus, A. Csiszárik, M. Matolcsi, D. Varga and P. Zsámboki, *The density of planar sets avoiding unit distances*, Math. Program. 207 (2024) 303–327; arXiv:2207.14179.

[2] C. Bachoc, A. Passuello and A. Thiery, *The density of sets avoiding distance* 1 *in Euclidean space*, Discrete Comput. Geom. 53 (2015) 783–808; arXiv:1401.6140.

[3] E. DeCorte, F. M. de Oliveira Filho and F. Vallentin, *Complete positivity and distance-avoiding sets*, Math. Program. 191 (2022) 487–558; arXiv:1804.09099.

[4] S. Galán, *Measurable density bounds hold for all colourings of the plane with algebraic forbidden distances*, preprint, version 8 (11 October 2026), `https://github.com/decalion89/chromatic-number-of-the-plane`.

[5] T. Keleti, M. Matolcsi, F. M. de Oliveira Filho and I. Z. Ruzsa, *Better bounds for planar sets avoiding unit distances*, Discrete Comput. Geom. 55 (2016) 642–661; arXiv:1501.00168.

[6] F. M. de Oliveira Filho and F. Vallentin, *Fourier analysis, linear programming, and densities of distance avoiding sets in* $\mathbb{R}^n$, J. Eur. Math. Soc. 12 (2010) 1417–1428; arXiv:0808.1822.

[7] L. A. Székely and N. C. Wormald, *Bounds on the measurable chromatic number of* $\mathbb{R}^n$, Discrete Math. 75 (1989) 343–372.